

EVALUATING FEATURE PREDICTION FOR 2D POSE TRAJECTORY RESTORATION WITH PAIRED SYNTHETIC SUPERVISION

Anonymous authors

Paper under double-blind review

ABSTRACT

Pose restoration must retain side-sensitive movement measurements as well as accurate trajectories. We evaluate reference-feature prediction using paired synthetic supervision and a signed difference in left and right knee excursion. Rendered motion combines controlled knee edits, physical mirroring, occlusion, and joint-label corruption. Three sequential experiments train on 112 people and reuse 14 development people across three seeds. Direct coordinate training reduces response error from 12.69° to 7.54° and waveform error from 18.57° to 12.07° . The original change-supervised objective worsens waveforms across eight model families. In the primary feature-difference repair setting on ViTPose, reducing scalar weight recovers 93% of the dense arm’s mean waveform gain. Feature-difference pre-training has a mean response advantage of 0.37° over endpoint-feature supervision (95% interval $[-1.11, 1.76]^\circ$). In a post hoc laterality analysis, this lower response point estimate coexists with higher mean geometric assignment failure (50.46% versus 49.69%). Failure costs materially affect response scores, and zero-response prediction beats all 16 neural variants in the pooled response experiment. The declared primary comparisons leave an incremental feature-prediction benefit unresolved; the complementary measurements identify trajectory and side-assignment errors that the scalar response alone does not capture.

1 INTRODUCTION

A plausible pose trajectory can alter the movement measurement extracted from it. When a video system compares walking across conditions, reconstruction may attenuate an angular change or assign it to the other leg. We study *laterality*, meaning anatomical left–right labeling, using a signed bilateral measurement and a geometric assignment diagnostic. The signed measurement alone does not identify which leg changed.

Predicting reference features may help retain movement structure in noisy observations. Joint-embedding predictive architectures (JEPAs) motivate this possibility (Assran et al., 2023; Bardes et al., 2024), but their objectives do not guarantee downstream angular accuracy. A supervised readout—a small network converting features to coordinates—can exploit observed-coordinate shortcuts, while a loss on one movement summary leaves much of the trajectory unconstrained.

We ask whether reference-feature prediction improves a controlled side-sensitive response beyond coordinate learning, and whether change supervision helps without degrading trajectories. The declared primary tests paired differences against endpoint supervision with the original change-supervised readout; coordinate-only readouts examine dependence on that choice. A subsequent post hoc assignment analysis examines whether lower response scores coincide with better geometric side labeling. Initialized and shuffled-reference controls assess pretraining’s role. The contribution is this controlled evaluation, its readout-weight follow-up, and failure accounting, within a compact restoration system using privileged synthetic targets and a repeatedly inspected development population.

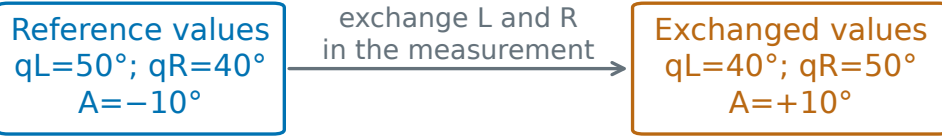
2 A LATERALITY-SENSITIVE RESTORATION TARGET

For each leg $\ell \in \{L, R\}$, let $\theta_{\ell,t}$ be the image-plane angle between knee-to-hip and knee-to-ankle vectors. We define angular excursion and its signed bilateral difference by

$$q_\ell = P_{95}(\theta_\ell) - P_5(\theta_\ell), \quad A = q_R - q_L, \quad \Delta A = A_b - A_a. \quad (1)$$

Here a and b are complete movement states of the same source window; ΔA is a between-state contrast, not a time derivative. Percentiles reduce dependence on isolated extremes. The primary response error is $E_\Delta = |\Delta \hat{A} - \Delta A|$, in degrees. A hat denotes a value computed from restored coordinates. Camera and observation conditions are held fixed within a movement contrast, and each camera has its own projected reference.

A Idealized exchange of side measurements



B A naming error changes the observation

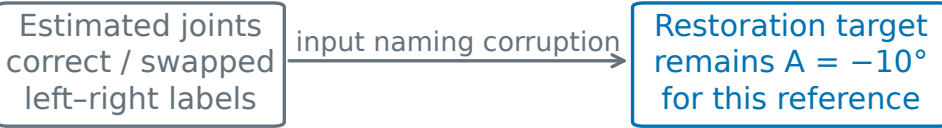


Figure 1: Why must physical mirroring and label corruption have different targets? The numerical excursions are an algebraic illustration, not experimental observations. Panel A exchanges already-computed left and right excursion values, which reverses A algebraically. The executed three-dimensional mirror is reprojected through a fixed camera and need not exchange projected excursions or reverse A . In panel B, a naming error changes only the estimated input; the anatomically named reference remains fixed. Solid arrows describe transformations, not learned model operations. This target distinction motivates the laterality-sensitive evaluation without imposing learned equivariance.

The distinction in Figure 1 prevents a left–right naming error from being interpreted as a movement response. Physical mirroring changes reference anatomy and observation together, with projected measurements recomputed for each camera. A global or temporary input-label exchange instead creates an observation error to be repaired. These transformations are crossed with movement edits; pooling them supplies a stress test, while condition-specific analysis is needed to diagnose its failures.

A favorable response score has important ambiguities. If $e_i = \hat{A}_i - A_i$, then $E_\Delta = |e_b - e_a|$: identical nonzero endpoint errors cancel. Different trajectories can also have the same percentile excursion. We therefore retain full knee-angle waveform error, normalized coordinate error, response-direction accuracy, and geometric left–right assignment failure. None can substitute for the others. Predicting zero response supplies an additional measurement baseline: it can score well when true projected changes are small, despite recovering no change.

Geometric assignment uses reference-valid hip, knee, and ankle pairs separated by at least four pixels. Let d_n and d_s be summed Euclidean errors under named and swapped assignment. A case is wrong if $d_s < d_n - 2$ pixels, ambiguous if $|d_n - d_s| \leq 2$ pixels, and missing if the predictions are nonfinite. All three count as failures. This denominator is independent of angular support, and a 50% failure score is not a calibrated chance level.

3 CONTROLLED MOTION AND OBSERVATION DATA

The source is AMASS (Mahmood et al., 2019), which brings recorded motion-capture collections into a common body-model representation. No new participants were recruited. Walking candidates are selected from the eligible manifest, divided into nonoverlapping windows, and screened for usable projected geometry. Filename-based selection and geometry screening do not establish clinical status or validate every retained interval as natural gait.

Each window contains 128 timestamps at 25 Hz, about 5.1 seconds. A local right-knee rotation is composed with the recorded motion, gated by an ankle-height proxy and tapered to vanish at window boundaries. Nominal edit levels are 0, 5, 10, and 15 degrees. The edit is applied before physical mirroring, which reflects geometry and exchanges anatomical sides. A nominal edit is a body-model operation; its magnitude need not equal the resulting projected ΔA . These interventions do not simulate disease or treatment response.

Rendering-derived person boxes are supplied to the pose estimators; detector uncertainty is outside this study. Cameras are fitted once across the original, edited, and mirrored geometries of a source window. Rendered images are processed by fixed RTMPose-M, HRNet-W32, and ViTPose-base estimators (Jiang et al., 2023; Sun et al., 2019; Xu et al., 2022). The restoration input includes bilateral shoulders, elbows, wrists, hips, knees, and ankles, for 12 two-dimensional joints. Projected body-model joints provide reference coordinates even when hidden in the image. Two cameras (45° oblique and 90° side), original and mirrored physical states, and clear or added-occlusion observations are crossed with correct, globally swapped, or temporarily swapped joint names. The temporary swap covers the middle third of a window. Coordinates, confidence, and observed indicators are swapped together; reference names remain fixed.

Table 1: Completed population and training exposure. Counts describe admitted people and source windows, before repeated rendering, estimator, and naming conditions. The development cohort is reused by all three stages. The protected-person confirmation has no completed result in this evidence packet.

Population	People	Windows	Role
Training	112	1,645	Fitting and calibration
Development	14	155	Reused evaluation

Training comprises 692 raw motions. Development includes 12 people from BioMotionLab_NTroje and two from KIT. Its 155 windows expand to 55,800 endpoint records, including a duplicated zero-level endpoint for a no-change diagnostic. Repeated renderings and endpoints do not add independent people. The 15° edit and ViTPose observations are excluded from restoration fitting and included in development evaluation. Generalization to these conditions is distinct from independent-population confirmation. GAVD supplies no examples to the completed experiments reported here.

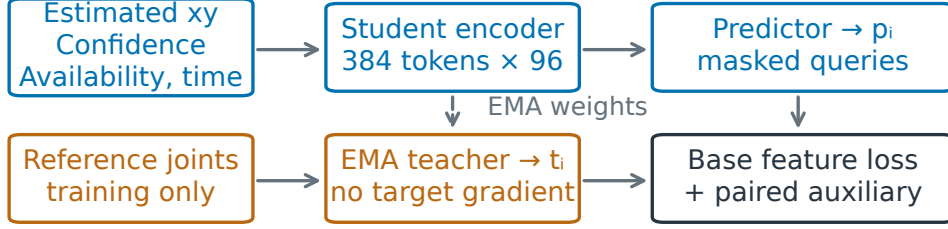
For retained observed points C , input normalization uses origin $o = \text{median}(X_C)$ and isotropic scale $s = \|P_{95}(X_C) - P_5(X_C)\|_2$, where quantiles act on coordinate dimensions. Fewer than two retained points, nonfinite scale, or $s < 10^{-6}$ use fixed $o = 0, s = 1$. Student inputs and reference targets share $(X - o)/s$; references and artificially hidden observations never determine this transform.

The reference is exact for the chosen synthetic projection, subject to the body model and geometry conventions; it is not an independent anatomical or clinical measurement. View-dependent 2D angles can change with camera placement even when 3D motion is unchanged. Systems such as OpenCap (Uhlrich et al., 2023) illustrate the importance of independent movement validation when converting video landmarks into biomechanical estimates. The present experiment addresses an earlier question: what information survives the restoration of controlled projected observations?

4 PREDICTIVE REPRESENTATIONS AND PAIRED SUPERVISION

Our S-JEPA-inspired predictor (Abdelfattah & Alahi, 2024) adapts skeletal action-recognition pre-training to estimated-input/reference-target restoration. A feature is a numerical vector representing a joint-time token. Four successive frames per joint give 384 tokens of width 96, processed by a four-layer, four-head transformer. Pretraining hides approximately half the available tokens in connected joint regions and time intervals. This graph-time mask is distinct from image occlusion; other implemented mask policies have no completed comparisons here.

A Predictive pretraining (paired states a and b)



B Single-state encoder and readout path

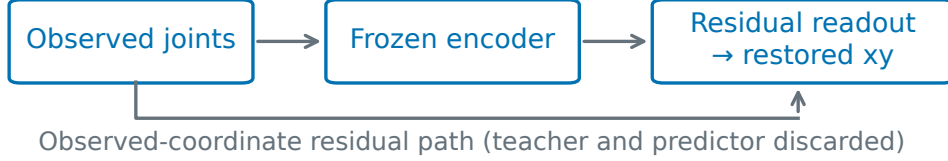


Figure 2: What is trained and deployed? Solid arrows carry data or targets; the dashed arrow updates the teacher by exponential moving average (EMA) of student weights. The teacher receives projected references in observation-derived normalization. Paired auxiliaries compare states a, b at common queries. The lower path computes one state; paired readout training compares two outputs, while inference needs one observed state. Teacher and predictor are discarded. Direct coordinate training instead adapts encoder and readout jointly.

Artificial masking zeros coordinate, confidence, and availability channels while retaining positions and timestamps. A two-layer predictor matches centered and sharpened teacher distributions with temperature-scaled cross-entropy (Caron et al., 2021), plus $0.05L_{\text{VICReg}}$ on projector-head outputs from translated-view encoder means. The regularizer weights invariance and variance by 25 each and covariance by one (Bardes et al., 2022). This differs from I-JEPA feature regression. The reference teacher is privileged; the full-window restorer has no forecasting or planning task.

For a common queried token of dimension $D = 96$, write $H(v) = v - \mathbf{1}(\sum_d v_d)/D$ and

$$e_i = H[p_i/0.1 - \text{sg}\{(t_i - c)/0.06\}], \quad L_\Delta = \frac{\|e_b - e_a\|^2}{2D}, \quad L_E = \frac{\|e_a\|^2 + \|e_b\|^2}{2D}. \quad (2)$$

Here p_i, t_i, c denote predictor output, teacher output, and running center; sg stops target gradients. Auxiliaries average common queries with all four reference frames valid in both states, first within pairs, then equally across pairs. Both retain the base loss and share $\lambda_J = 0.1\sqrt{G_{\text{base}}/\max(G_\Delta, G_E)}$, where G sums squared gradients over all trainable parameters on 32 training batches at seed-17 initialization; this coefficient is reused across seeds. On identical tensors, $L_\Delta = L_E - e_a^\top e_b/D$, allowing shared residuals to cancel. Separately evolving teachers limit mechanistic attribution. A coordinate-difference control converts residuals to a common observation-derived scale before differencing.

5 READOUTS, EXPERIMENTAL DESIGN, AND INFERENCE

The readout is a small network converting encoder features into coordinate corrections. It adds a predicted correction at observed positions and predicts missing coordinates directly. Frozen initialized features and a shuffled-reference JEPA control test whether useful restoration requires learned observation–reference correspondence. Shuffling assigns a different accepted window from the same person while retaining endpoint role and observation conditions. It does not randomize the paired movement states in the new feature-difference auxiliary.

The base readout minimizes coordinate error L_{coord} . The original change objective adds $L_{\text{scalar}} = \mathbb{E}[(\Delta\hat{A} - \Delta A)/180]^2$ and a short-segment geometry penalty L_{geom} , both at unit weight. Comparing it with base therefore tests a package of two additions. The repair experiment keeps $L_{\text{coord}} + L_{\text{geom}}$ fixed and compares $0.1L_{\text{scalar}}$ against a calibrated dense term

$$L_{\text{dense}} = \mathbb{E}_{(a,b)} \left[\frac{1}{|S_{ab}|} \sum_{(\ell,t) \in S_{ab}} \left[\frac{(\hat{\theta}_{b,\ell,t} - \hat{\theta}_{a,\ell,t}) - (\theta_{b,\ell,t} - \theta_{a,\ell,t})}{180} \right]^2 \right]. \quad (3)$$

S_{ab} is common reference support for a pair; supported pairs receive equal weight. The new term constrains change at each leg and timestamp rather than only its excursion summary. Its initial auxiliary gradient energy is matched to the low-scalar arm using 32 training-only batches for each encoder and seed. Writing G_s and G_d for sums of squared readout-gradient norms over those batches gives $\lambda_d = 0.1\sqrt{G_s/G_d}$. This controls initial scale, not the complete optimization trajectory.

Table 2: Completed sequential experiments. Each uses seeds 17, 29, and 43. Final counts include the three seeds, not three independent populations. Core baselines also include six deterministic controls.

Stage	Controlled question	Pretrain	Final
Core	Five families, each with base and original change objectives	9	30
Response	Delta features, endpoint features, coordinate differences; two readouts each	9	18
Repair	Delta and endpoint encoders; dense versus low scalar	0	12

Pretraining and readout fitting each use 2,000 updates; direct end-to-end fitting uses 4,000. Batch size is 16. AdamW uses peak learning rate 3×10^{-4} , weight decay 0.01, 5% warmup followed by cosine decay, and gradient-norm clipping at one. Equal update totals do not match coordinate-supervised exposure or encoder adaptation. Training samples hierarchically by person, motion, window, and pair. No broad coefficient or convergence study is available.

Evaluation fixes reference support before examining predictions, using a common intersection across source-family variants. At least 16 frames and 80% coverage must satisfy finite-angle and two-pixel segment checks. Nonfinite predictions or predicted segments below two pixels invalidate the measurement on admitted support. Failures remain in the denominator, with waveform cost 180° and response cost 720° . Coordinate error divides joint distance by the rendered person-box diagonal. Direction accuracy requires $|\Delta A| > 1^\circ$ and counts failures as wrong.

Means average conditions within windows, windows within motions, motions within people, and seeds. Core and response primary intervals use 2,000 crossed bootstrap draws, resampling people and seeds independently while preserving method pairing (seed 731). Repair’s declared interval first averages seeds within each person and uses a paired-person t interval over 14 people; it is conditional on the fitted seeds. Condition reaggregation weights nonheld versus held levels 4:1 for endpoint metrics (including the duplicated zero endpoint), and 2:1 for nonzero responses. All are development estimates. Later stages followed inspection of earlier results, and secondary comparisons are unadjusted for multiplicity.

6 RESULTS

6.1 COORDINATE RESTORATION HELPS; THE ORIGINAL CHANGE OBJECTIVE DAMAGES TRAJECTORIES

Direct coordinate fitting reduces response error from 12.69° for unchanged observations to 7.54° , waveform error from 18.57° to 12.07° , and all-joint normalized landmark error (NLE) from 0.0718 to 0.0298. The exploratory response improvement is 5.14° with crossed 95% interval $[2.15, 8.65]^\circ$. Every person’s seed-averaged waveform and coordinate scores improve, whereas response improves for 10 of 14 people.

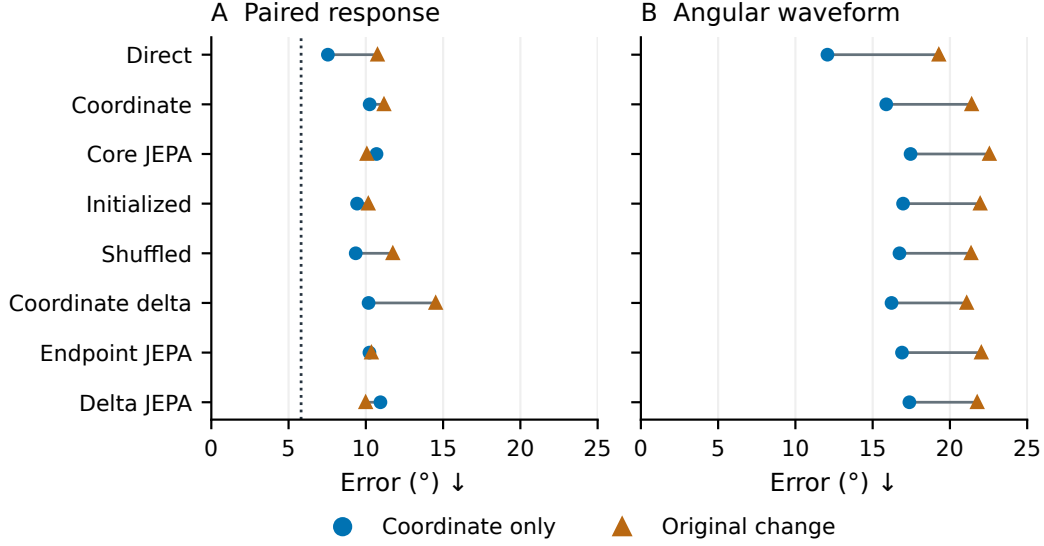


Figure 3: How does the original change objective affect response and trajectory error? Each line joins two readouts within a model family: blue circles use coordinate supervision and orange triangles add the original scalar-change and geometry terms. Points are descriptive hierarchical means over the same 14 development people and three seeds, pooled over estimators. All eight waveform means deteriorate, while response effects vary. The dotted line in A is zero-response prediction (5.81°), which has no corresponding waveform. Scores include failures; no uncertainty bars are implied.

The core shows waveform deterioration for all five families, every person’s seed-averaged score, and every seed’s population mean. Mean increases span $4.63\text{--}7.21^\circ$. The three response families extend the mean pattern to eight, with a minimum increase of 4.39° . These correlated comparisons describe one development population, rather than eight replications. Core paired JEPA illustrates why response alone is insufficient: its score improves from 10.69° to 10.07° , while waveform error rises from 17.45° to 22.55° .

Table 3: Pooled descriptive means over 14 people and three seeds. Base uses coordinate supervision; change adds scalar and geometry losses. NLE is dimensionless all-valid-joint error. Lower is better; statistical ordering is not established.

Method / objective	Response ($^\circ$)	Waveform ($^\circ$)	NLE
Unchanged observations	12.69	18.57	.0718
Direct / base	7.54	12.07	.0298
Initialized / base	9.43	16.97	.0687
Shuffled JEPA / base	9.35	16.74	.0676
Coordinate pretraining / base	10.25	15.88	.0487
Endpoint JEPA / change	10.36	22.03	.0704
Delta JEPA / change	9.99	21.76	.0699

6.2 PRIMARY CONTRASTS REMAIN UNCERTAIN, AND FAILURES AFFECT THEIR INTERPRETATION

The declared core primary compares JEPA with direct fitting under the original change objective, not the stronger direct/base model. Its response gain is 0.69° $[-0.64, 1.98]^\circ$. The response follow-up compares delta with endpoint JEPA under that same readout: 9.99° versus 10.36° , a gain of 0.37° $[-1.11, 1.76]^\circ$. Neither establishes an incremental benefit. The corresponding base-readout means reverse the ordering, 10.94° versus 10.23° . The secondary interaction is 1.09° $[-0.65, 2.85]^\circ$, leaving the apparent readout dependence unresolved.

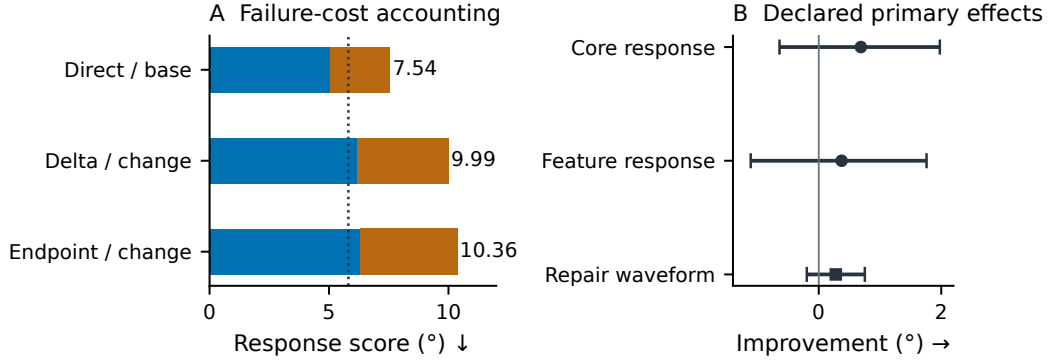


Figure 4: What supports the apparent gains? A: response error separates into successful-output contribution (solid blue) and failure-cost contribution (hatched orange) at identical population weights. The dotted line is zero-response prediction. The successful contribution assigns zero to failures and is not a conditional success-only mean. B: comparator-minus-candidate effects; positive favors the candidate. Core compares paired JEPA with direct fitting, and feature response compares delta with endpoint JEPA, both under the original change readout. Circles show crossed people/seed 95% intervals pooled over estimators. Repair compares delta/dense with delta/low-scalar on ViTPose waveform error; its square shows a person- t 95% interval conditional on three seeds. The rows have different estimands and cannot be pooled. All use 14 people.

Writing F for prediction failure and w for the hierarchical weights, the response score decomposes as

$$\bar{E}_\Delta = \mathbb{E}_w[\mathbf{1}_{\neg F}|\Delta\hat{A} - \Delta A|] + 720 \Pr_w(F). \quad (4)$$

Delta and endpoint JEPA have failure contributions 3.778° and 4.055° and successful contributions 6.210° and 6.306° . Thus 0.277° , about 74% of their 0.373° mean contrast, comes from the penalty term. This is score accounting, not a causal decomposition; the total remains uncertain. Direct/base has contributions 2.475° and 5.069° , respectively.

Zero-response prediction achieves 5.811° , below all 16 neural variants in the response export. Improving on noisy input therefore does not establish useful recovery under the pooled metric. Direct/base still contains response information: direction accuracy is 66.1%, and clear-observation response error is 3.90° versus 11.19° under occlusion. Zero prediction scores 5.81° in both strata. These descriptive results identify observation sensitivity without establishing natural-video calibration. A 50% direction score would not itself establish chance performance without the sign distribution and a corresponding baseline.

6.3 LOWER SCALAR WEIGHT RECOVERS MOST MEAN WAVEFORM IMPROVEMENT

On held-out-estimator ViTPose observations, delta JEPA’s waveform error falls from 23.17° with the original change objective to 19.47° with its scalar coefficient reduced tenfold, and 19.19° with dense angular-change supervision. Reducing the weight recovers about 93% of the dense arm’s mean improvement over the original objective. This ratio describes mean gains and does not allocate causal credit.

Dense versus low scalar has primary waveform gain 0.28° [−0.19, 0.75]° under the person-*t* interval; a crossed-bootstrap sensitivity interval is [−0.44, 0.86]°. Its response gain is 0.13° [−2.69, 2.94]°. These intervals establish neither superiority nor preserved response accuracy. Endpoint JEPA has a favorable secondary waveform gain of 0.72° [0.39, 1.05]°; it is unadjusted and cannot replace the delta primary. Both dense arms remain roughly six degrees above direct/base waveform error on ViTPose.

6.4 A LOWER RESPONSE SCORE DOES NOT ESTABLISH GEOMETRIC SIDE RECOVERY

A post hoc analysis separates naming conditions in the completed exports (Figure 5). Direct/base has lower assignment failure than either predictive variant, especially under global swapping. This is a practical comparison: direct training updates the encoder with coordinate supervision, while JEPA encoders remain frozen during readout fitting.

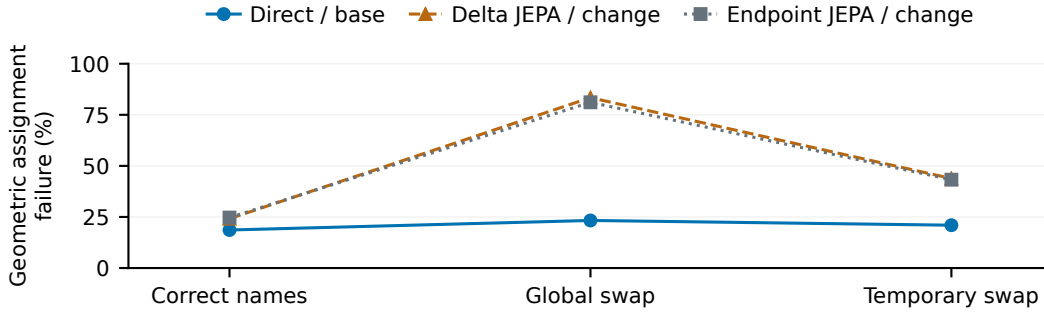


Figure 5: Which naming conditions remain difficult? Geometric assignment failure on reference-valid hip, knee, and ankle pairs separated by at least four pixels. Wrong, ambiguous (within two pixels of the alternative assignment), and missing predictions count as failures. Points are descriptive hierarchical means over 14 development people and three fitted seeds, pooled over cameras, physical orientations, observation conditions, and all three estimators. The post hoc analysis uses 4:1 nonheld/held endpoint weights; no error bars or chance baseline are implied. Global naming corruption is substantially more difficult for both frozen JEPA variants than for direct/base.

Delta/change has 50.46% assignment failure overall, versus 49.69% for endpoint/change, despite its lower response-error point estimate. The exploratory paired difference is 0.77 percentage points against delta (95% crossed interval [0.17,1.53]); it is unadjusted and cannot replace the response primary. Direct is not best on every outcome: its left/right excursion errors, including 180° failure costs, are 17.94/17.19°, versus 15.70/15.29° for delta/change. These results distinguish response, trajectory, and side-assignment fidelity rather than establish a universal ranking.

7 DISCUSSION, REPRODUCIBILITY, AND LIMITATIONS

The experiments support a bounded scientific claim: evaluating reference-feature prediction for restoration requires separating a signed movement response from trajectory quality and prediction reliability. Direct coordinate training improves the observed poses under this protocol. The original change-supervised package can reduce a scalar response error while worsening its supporting trajectory, and lowering the scalar weight recovers 93% of the dense arm’s mean waveform improvement in the primary repair setting. The uncertain primary contrasts leave the incremental value of the tested feature-prediction modifications unresolved.

The laterality experiment uses augmentation and anatomically labeled supervision. It includes no architectural equivariance constraint or completed direct equivariance test.

S-JEPA (Abdelfattah & Alahi, 2024) and general feature prediction in masked skeleton modeling (Sun et al., 2025) already investigate skeletal latent targets for action recognition. S-JEPA uses its target encoder downstream; our deployed encoder processes estimated coordinates, while reference coordinates supply privileged teacher targets during training. PoseBERT (Baradel et al., 2022) learns temporal pose reconstruction from motion-capture data, and MotionBERT (Zhu et al., 2023) learns motion representations through masked and noisy 2D-to-3D lifting. Our contribution is the controlled evaluation of a signed paired response, its geometric side assignment, and its downstream loss tradeoffs. Finite paired differences also differ from derivative supervision in Sobolev training (Czarnecki et al., 2017). These works and SmoothNet (Zeng et al., 2022) were not fitted as external baselines, so the results characterize only the implemented procedures.

The empirical limitations are substantial. Fourteen development people from two collections provide modest independent evidence despite many rendered conditions. The same cohort informed the sequential experiments, and three seeds give coarse information about training variability. There is no completed protected-person confirmation, GAVD evaluation, or clinical reference study. Two-dimensional angles are view-dependent, and synthetic edits do not reproduce natural pathology or acquisition errors. The zero-response result further limits any claim of reliable measurement recovery.

Optimization also constrains interpretation. The fixed budgets, weights, and compact residual read-out leave convergence and hyperparameter sensitivity unresolved. At calibration, the shared feature coefficient sets the larger endpoint auxiliary to 10% of base gradient magnitude, while the weighted delta auxiliary is 0.0013%. Equal coefficients therefore do not match initial gradient influence; these values do not determine later influence. Gradients in the six new feature-prediction pretraining fits were clipped on nearly every update. Initial gradient matching does not keep subsequent gradients matched. Dense supervision changes temporal information and gradient locations, so the repair cannot identify percentile-gradient sparsity as its sole mechanism. Separate teachers and the absence of a new movement-pair randomization control prevent unique attribution to feature-difference coupling.

For reproducibility, the study retains configuration, fitting ledgers, person-level scores, condition-level exports, comparisons, and diagnostic receipts for each stage. The accompanying audit links every numerical claim and vector plot to these artifacts; raw-data training is distinct from recomputing published summaries. The local transferred packet does not include raw rendered frames, per-example coordinate reconstructions, or complete checkpoints, so it cannot support a verified qualitative reconstruction panel or an end-to-end rerun on its own. Tutorial fixtures demonstrate calculations and are excluded from scientific results.

A decisive next study would freeze the response and waveform comparisons before evaluating unspected people, retain zero-response and direct-coordinate controls, and vary loss weight and training budget. It would separately assess anatomical naming under natural observations with synchronized references. The present evidence characterizes measurement and objective sensitivity under paired synthetic supervision, with representation benefit still unresolved.

AI USE DISCLOSURE

AI assistants helped inspect source code, notebooks, and saved experiment exports; formulate and critique the scientific framing; search primary literature; draft and revise the manuscript; create plotting and document-build code; and check numerical summaries and rendered pages. Independent assistant review considered scientific scope, evidence, statistics, and writing. The assistants did not collect participants, run the source training experiments, or provide independent empirical validation. Human authors remain responsible for verifying the complete manuscript, references, disclosure, and submission metadata before submission; this draft does not assert that such verification has occurred.

REFERENCES

- Mohamed Abdelfattah and Alexandre Alahi. S-JEPA: A joint embedding predictive architecture for skeletal action recognition. In *European Conference on Computer Vision*, pp. 367–384, 2024. URL https://www.ecva.net/papers/eccv_2024/papers_ECCV/papers/04755.pdf.
- Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 15619–15629, 2023. URL https://openaccess.thecvf.com/content/CVPR2023/html/Assran_Self-Supervised_Learning_From_Images_With_a_Joint-Embedding_Predictive_Architecture_CVPR_2023_paper.html.
- Fabien Baradel, Romain Brégier, Thibault Groueix, Philippe Weinzaepfel, Yannis Kalantidis, and Grégory Rogez. PoseBERT: A generic transformer module for temporal 3D human modeling. *arXiv preprint arXiv:2208.10211*, 2022. URL <https://arxiv.org/abs/2208.10211>.
- Adrien Bardes, Jean Ponce, and Yann LeCun. VICReg: Variance-invariance-covariance regularization for self-supervised learning. In *International Conference on Learning Representations*, 2022. URL <https://openreview.net/forum?id=iWpcWZ8phD>.
- Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. Revisiting feature prediction for learning visual representations from video. *arXiv preprint arXiv:2404.08471*, 2024. URL <https://arxiv.org/abs/2404.08471>.
- Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 9650–9660, 2021. URL https://openaccess.thecvf.com/content/ICCV2021/html/Caron_Emerging_Properties_in_Self-Supervised_Vision_Transformers_ICCV_2021_paper.html.
- Wojciech Marian Czarnecki, Simon Osindero, Max Jaderberg, Grzegorz Świrszcz, and Razvan Pascanu. Sobolev training for neural networks. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://papers.neurips.cc/paper_files/paper/2017/hash/758a06618c69880a6cee5314ee42d52f-Abstract.html.
- Tao Jiang, Peng Lu, Li Zhang, Ningsheng Ma, Rui Han, Chengqi Lyu, Yining Li, and Kai Chen. RTMPose: Real-time multi-person pose estimation based on MMPose. *arXiv preprint arXiv:2303.07399*, 2023. URL <https://arxiv.org/abs/2303.07399>.
- Naureen Mahmood, Nima Ghorbani, Nikolaus F. Troje, Gerard Pons-Moll, and Michael J. Black. AMASS: Archive of motion capture as surface shapes. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 5442–5451, 2019. URL https://openaccess.thecvf.com/content_ICCV_2019/html/Mahmood_AMASS_Archive_of_Motion_Capture_As_Surface_Shapes_ICCV_2019_paper.html.

-
- Ke Sun, Bin Xiao, Dong Liu, and Jingdong Wang. Deep high-resolution representation learning for human pose estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 5693–5703, 2019. URL https://openaccess.thecvf.com/content_CVPR_2019/html/Sun_Deep_High-Resolution_Representation_Learning_for_Human_Pose_Estimation_CVPR_2019_paper.html.
- Shengkai Sun, Zefan Zhang, Jianfeng Dong, Zhiyong Cheng, Xiaojun Chang, and Meng Wang. Towards efficient general feature prediction in masked skeleton modeling. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 12212–12221, 2025. URL <https://arxiv.org/abs/2509.03609>.
- Scott D. Uhlich, Antoine Falisse, Łukasz Kidziński, Julie Muccini, Michael Ko, Akshay S. Chaudhari, Jennifer L. Hicks, and Scott L. Delp. OpenCap: Human movement dynamics from smartphone videos. *PLOS Computational Biology*, 19(10):e1011462, 2023. doi: 10.1371/journal.pcbi.1011462. URL <https://journals.plos.org/ploscompbiol/article?id=10.1371/journal.pcbi.1011462>.
- Yufei Xu, Jing Zhang, Qiming Zhang, and Dacheng Tao. ViTPose: Simple vision transformer baselines for human pose estimation. In *Advances in Neural Information Processing Systems*, volume 35, 2022. URL https://proceedings.neurips.cc/paper_files/paper/2022/hash/fbb10d319d44f8c3b4720873e4177c65-Abstract-Conference.html.
- Ailing Zeng, Lei Yang, Xuan Ju, Jiefeng Li, Jianyi Wang, and Qiang Xu. SmoothNet: A plug-and-play network for refining human poses in videos. In *European Conference on Computer Vision*, 2022. URL https://www.ecva.net/papers/eccv_2022/papers_ECCV/papers/136650615.pdf.
- Wentao Zhu, Xiaoxuan Ma, Zhaoyang Liu, Libin Liu, Wayne Wu, and Yizhou Wang. MotionBERT: A unified perspective on learning human motion representations. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 15085–15099, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Zhu_MotionBERT_A_Unified_Perspective_on_Learning_Human_Motion_Representations_ICCV_2023_paper.html.